COURSE LABS · 2025–2026

Four labs at a glance

The question, one graph, the numbers, and what I concluded. Graphs rebuilt from my submitted reports' data; the workbook behind them recalculates every result.

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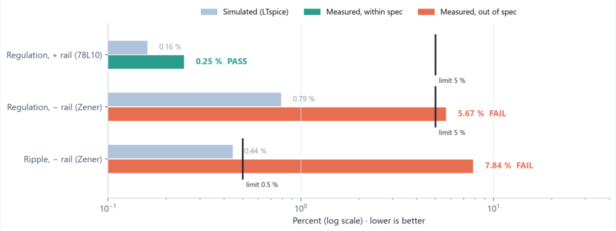
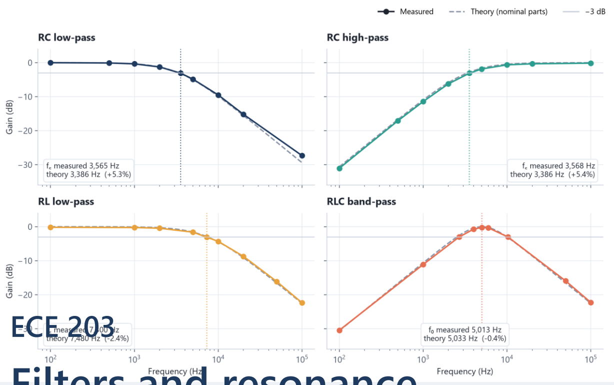


ECE 302

±10 V dual-rail power supply

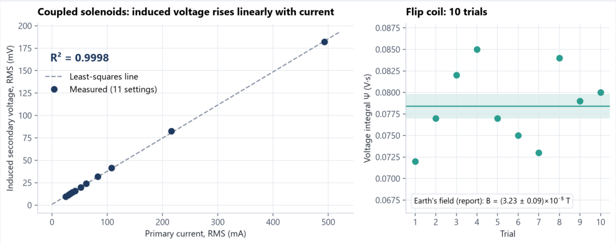
Design, simulate in LTspice, build, measure



ECE 203

Filters and resonance

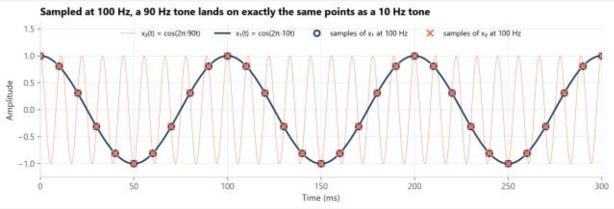
Four filters swept 100 Hz to 100 kHz



PHYS 230

Faraday's law

Earth's field from a flip coil, with error propagation



ECE 340

Sampling and aliasing

MATLAB: when 90 Hz becomes 10 Hz

ECE 302 · ELECTRONIC DEVICES · LAB 2

A ±10 V supply: one rail passed, one didn't, and why

THE QUESTION

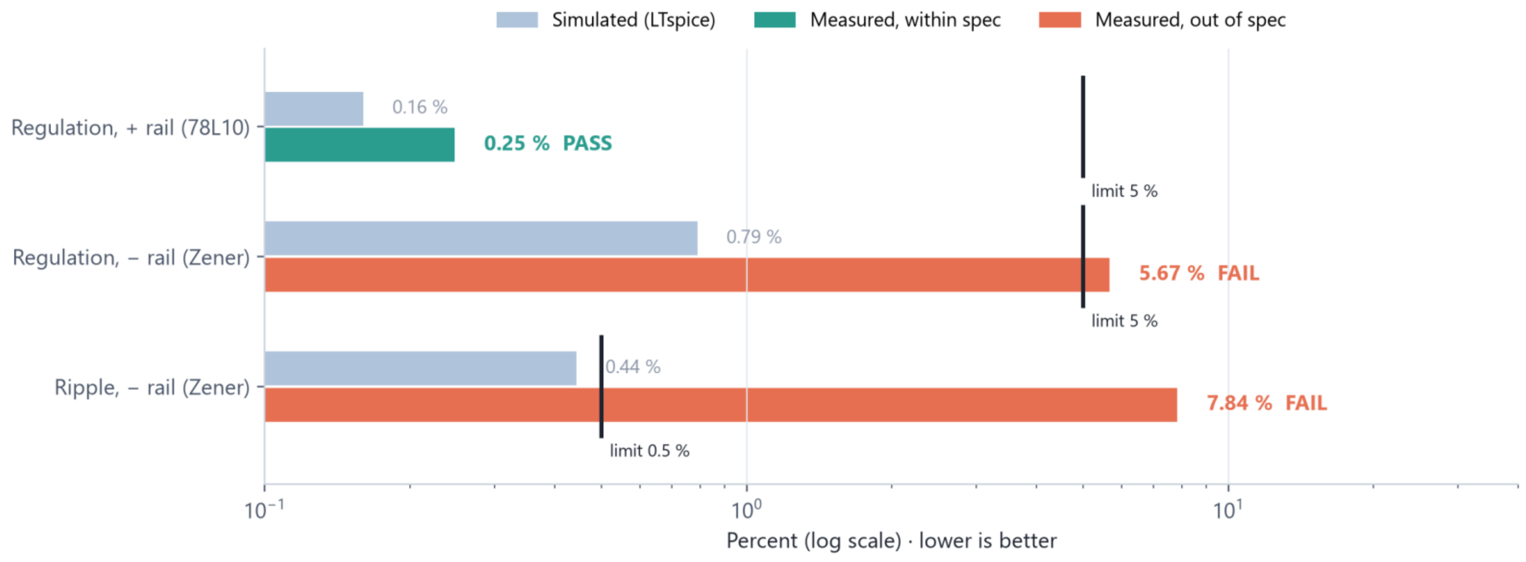
Can a transformer, rectifiers, a 78L10 regulator and a Zener shunt hold ±10 V within 5 % regulation and 0.5 % ripple at full load?

WHAT WE DID

Sized the 220 Ω ballast by hand, simulated both rails in LTspice, built it on a breadboard and measured with a scope and meter.

CONCLUSION

The IC regulator rail met spec. The Zener rail fell out of breakdown under heavy load: not enough ballast current. Fix: more filter capacitance or a smaller ballast.



0.25 %

+ rail regulation, measured (limit 5 %)

5.67 %

− rail regulation, measured: over the limit

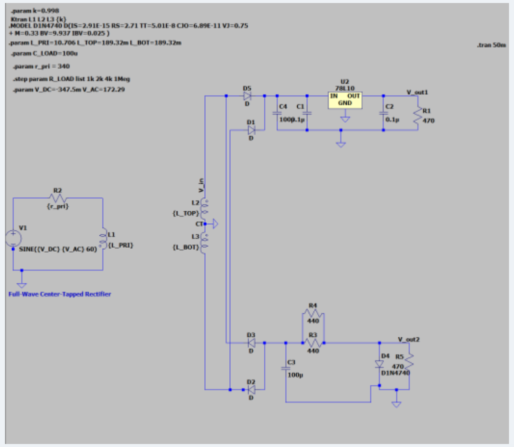
7.84 %

− rail ripple, measured (sim 0.44 %, limit 0.5 %)

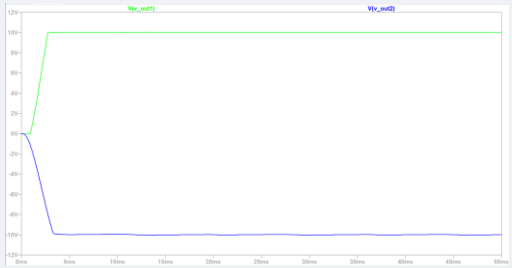
Group lab with Rohan Singh. Values from our submitted report (Oct 2026); chart rebuilt from them.

ECE 302 · LAB 2 · HOW IT WAS BUILT

From schematic to scope


.param k=0.998
Ktran L1 L2 L3 {k}
.MODEL D1N4740 D(IS=2.91E-15 RS=2.71 TT=5.01E-8 CJO=6.89E-11 VJ=0.75
+ M=0.33 BV=9.937 IBV=0.025)
.param L_PRI=10.706 L_TOP=189.32m L_BOT=189.32m
.param C_LOAD=100u
.param r_pri = 340
.step param R_LOAD list 1k 2k 4k 1Meg
.param V_DC=-347.5m V_AC=172.29
.tran 50m
D5
D
D1
D
C4
100µ
C1
0.1µ
U2
78L10
IN
OUT
GND
C2
0.1µ
V_out1
R1
470
R2
{r_pri}
V1
SINE({V_DC} {V_AC} 60)
L1
{L_PRI}
V_in
L2
{L_TOP}
CT
L3
{L_BOT}
Full-Wave Center-Tapped Rectifier
R4
440
R3
440
D3
D
C3
100µ
D2
D
V_out2
D4
D1N4740
R5
470

›



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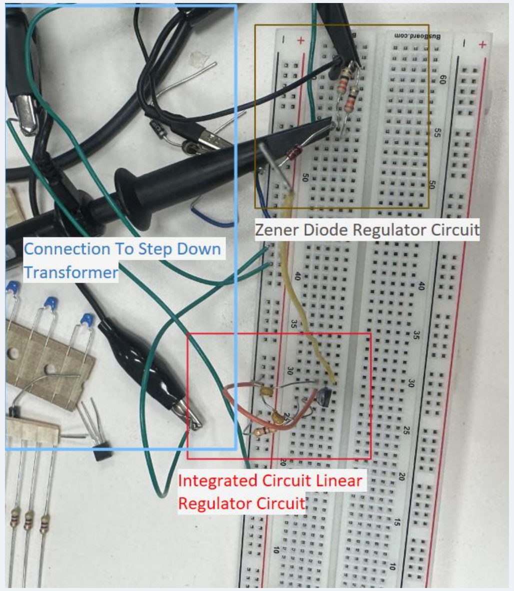
Zener Diode Regulator Circuit
Connection To Step Down Transformer
Integrated Circuit Linear Regulator Circuit

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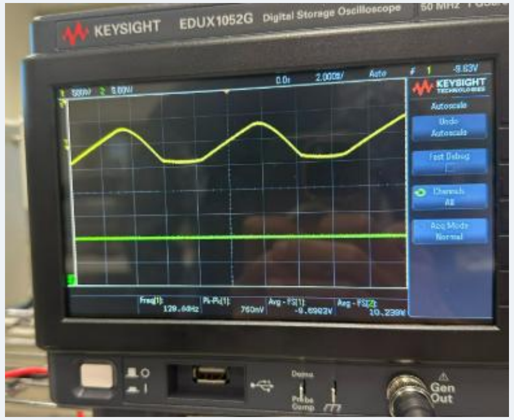

KEYSIGHT
EDUX1052G Digital Storage Oscilloscope
50 MHz
2.000s/
Auto
-9.63V
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels All
Acq Mode Normal
Freq(1):
120.00Hz
Pk-Pk(1):
760mV
Avg - FS(1):
-9.6902V
Avg - FS(2):
10.238V
Probe Comp
Gen Out

1 · Design in LTspice

Transformer, full-wave centre-tapped rectifiers, 78L10 on the + rail, 1N4740 Zener shunt on the − rail.

2 · Simulate

Transient run: both rails settle to ±10 V within about 3 ms. Load stepped with .step to get full-load and no-load.

3 · Build

On the breadboard: transformer leads in, IC regulator rail and Zener rail side by side.

4 · Measure

Scope in AC coupling on the − rail: 760 mV peak-to-peak ripple, the number that failed the spec.

Group lab with Rohan Singh. Figures from our submitted report: LTspice schematic and transient run, the breadboard build, the bench scope.

ECE 203 · ELECTRICAL CIRCUITS II · LAB 6

Four filters, measured against the math

THE QUESTION

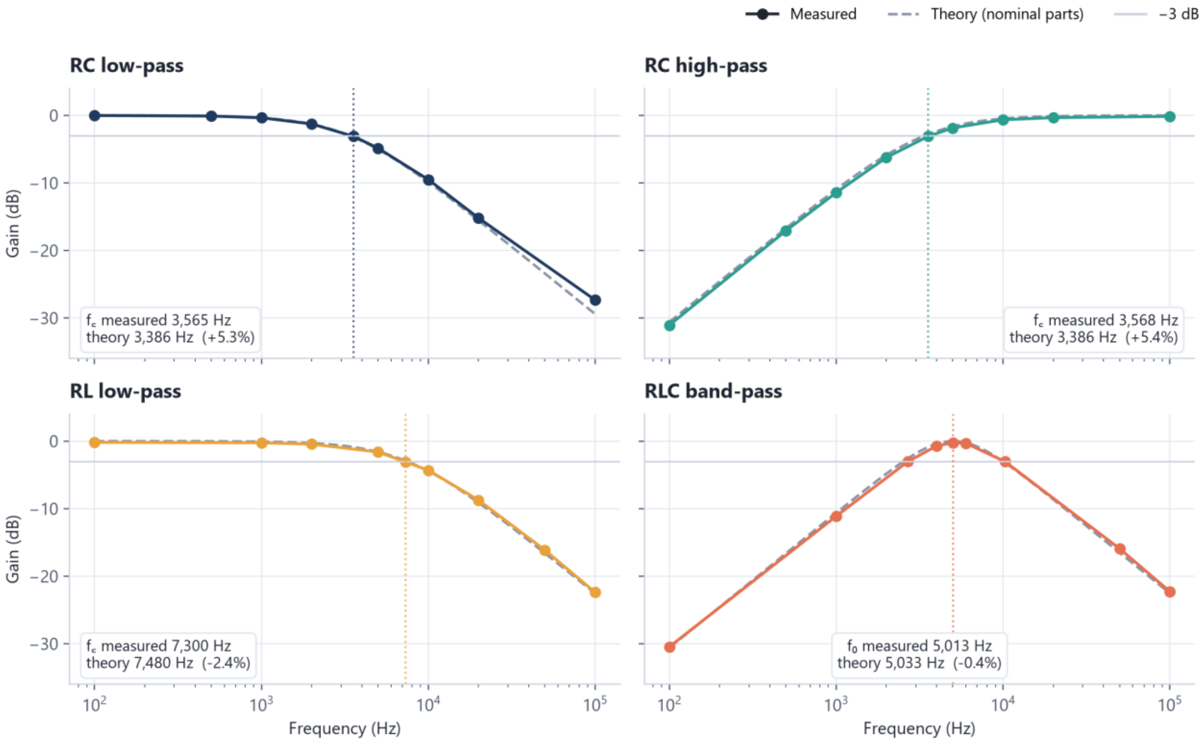
Do real RC, RL and RLC filters cut off where 1/(2πRC), R/(2πL) and 1/(2π√LC) say they should?

WHAT I DID

Swept each filter from 100 Hz to 100 kHz on the function generator and scope, and found each −3 dB point.

CONCLUSION

Every corner landed within 5.4 % of theory. Both RC cut-offs ran high by the same 5 %, consistent with the capacitor being about 95 nF (within its tolerance).



−0.4 %

RLC centre frequency vs theory (5,013 vs 5,033 Hz)

+5.3 %

Both RC cut-offs high: implies C ≈ 95 nF

Q 0.66

Band-pass sharpness, theory 0.67

Individual results sheet (Mar 2026). Measured points from my tables; theory uses the nominal 470 Ω, 100 nF, 10 mH.

Faraday's law, two ways

THE QUESTION

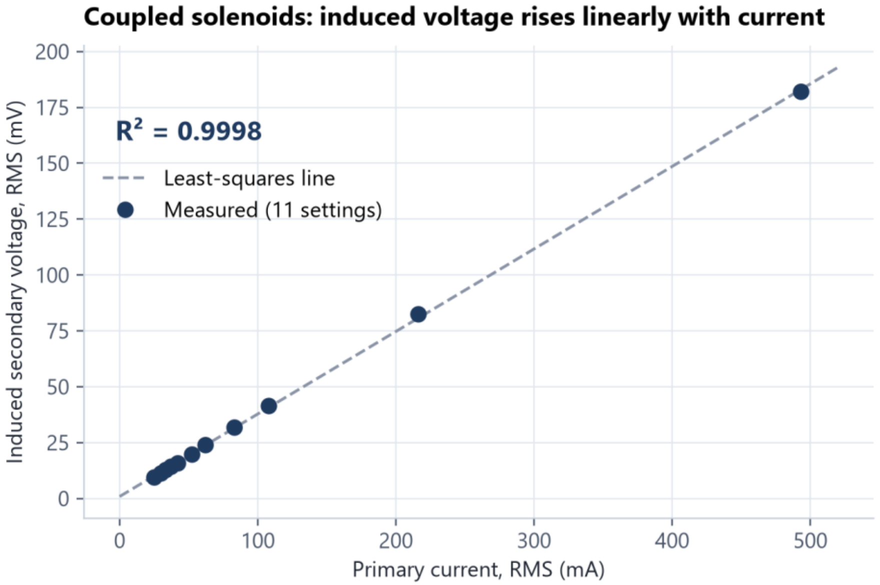
Is induced EMF proportional to the rate of change of flux, and can a flipped coil measure Earth's magnetic field?

WHAT I DID

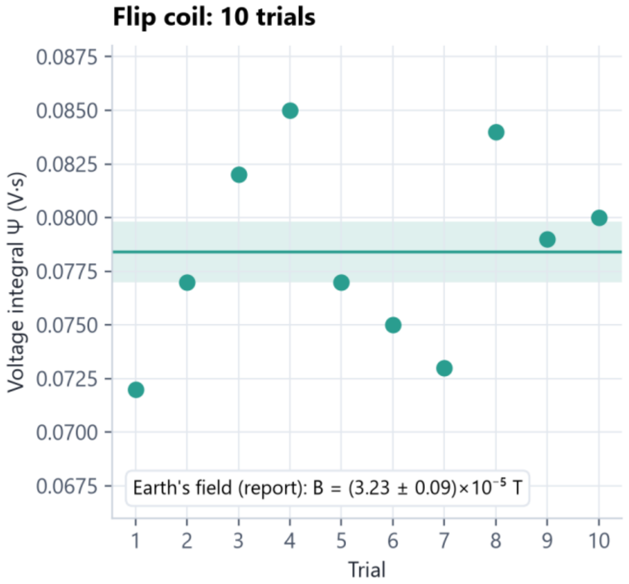
Ten 180° flips of a 100-turn coil through a ×200 amplifier; then 11 current settings on coupled solenoids, fit with LINEST in Excel.

CONCLUSION

Induced voltage is linear in current (R^2 = 0.9998). The flip coil gave B 32 % above the gaussmeter; likely local interference and coil alignment.



Coupled solenoids: induced voltage rises linearly with current



Flip coil: 10 trials

R^2 0.9998

Linear fit, induced EMF vs primary current

3.23 ± 0.09

Earth's field, ×10^{-5} T (fractional error propagation)

2.45

Gaussmeter reference, ×10^{-5} T

With lab partners Duong Hung Nguyen and James Pacia. Report written individually (Mar 2026).

When 90 Hz turns into 10 Hz

THE QUESTION

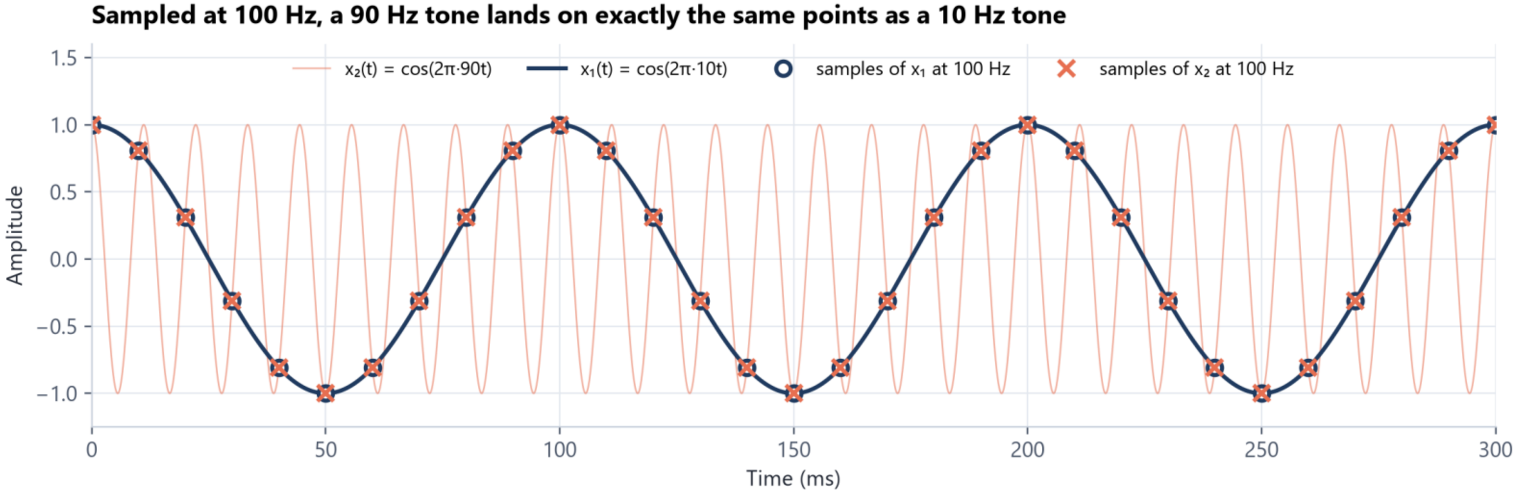
What happens when a signal is sampled slower than twice its highest frequency, in 1D and in images?

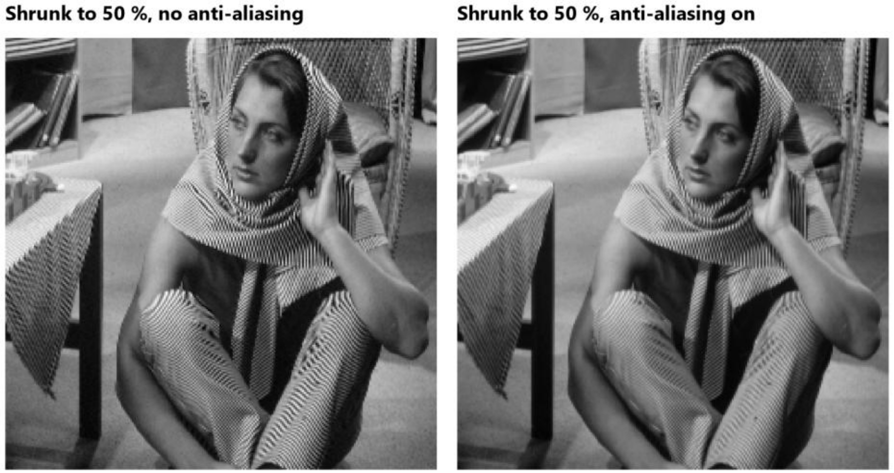
WHAT I DID

Sampled 10 Hz and 90 Hz cosines at 100 Hz and 1 kHz in MATLAB, then shrank a test image with anti-aliasing off and on.

CONCLUSION

At 100 Hz the two tones give identical samples (max difference 1.7×10^{-14}). Shrinking an image is resampling too: fine stripes alias into moiré unless low-passed first.


Sampled at 100 Hz, a 90 Hz tone lands on exactly the same points as a 10 Hz tone


Shrunk to 50 %, no anti-aliasing

Shrunk to 50 %, anti-aliasing on

Left: stripes on the clothing turn into false wavy bands.

Right: filtered first, they fade to grey instead. Cost: a little blur.

Individual report (Oct 2026). Top graph redrawn from the lab's signal definitions; the images are my MATLAB output.

WHAT THESE SHOW

Predict it, measure it, explain the gap.

Bench

Oscilloscope (AC/DC coupling), function generator, DMMs, breadboard builds

Simulation

LTspice: stepped parameters, .meas averages and ripple

Analysis

Excel LINEST, error propagation, MATLAB conv, sampling, imresize

Reporting

Spec checks, honest failures and their causes, Word reports

Data and every calculation: Lab_Highlights_Data.xlsx · anmol-portfolio-omega.vercel.app/docs